Supporting Information: Objective-Dependent Active Learning and Calibrated Uncertainty for Sample-Efficient Discovery of Rare-Earth Extractants

Krishna Harish*

Lawrence E. Elkins High School, Missouri City, Texas, USA

E-mail: krishnaharish2009@gmail.com

1 Datasets

f-element extraction (primary). 1,202 experimental lanthanide distribution coefficients ($\log D$) from Liu et al. (1,085 training + 117 validation), with 2,291 features per record: ligand ECFP/Morgan bits and RDKit physicochemical descriptors, experimental conditions (ligand and acid concentration, diluent identity and volume ratio, temperature), and lanthanide-identity descriptors. Target range $\log D \in [-4.00, 3.59]$. A strictly prospective external set of four subsequently synthesised ligands (56 ligand$\times$metal measurements) is used only for final validation.

Lipophilicity (generality). 4,200 octanol–water $\log D$ values (MoleculeNet). Features: 1,024-bit Morgan fingerprints (radius 2) concatenated with the RDKit physicochemical descriptors above.

2 Methods

Surrogate. Random forest, 300 trees, minimum 2 samples per leaf, features standardised. Epistemic uncertainty $\hat{\sigma}$ is the per-point standard deviation across trees.

Conformal calibration. Split-conformal: 20% of the training data held out as a calibration set; nonconformity scores $s_i = |y_i - \hat{\mu}_i|/\hat{\sigma}_i$; the calibrated interval half-width is $q_\alpha \hat{\sigma}$ with q_α the empirical α-quantile of $\{s_i\}$.

Active-learning protocol. Fixed 80/20 pool/test split. f-element: seed = 25, batch = 15, budget = 420. Lipophilicity: seed = 40, batch = 25, budget = 715. Six random seeds per strategy; curves report mean±s.d. Top-10 recall is the fraction of the ten highest-$\log D$ pool members acquired.

3 Full results

Table 1: Active learning on the f-element $\log D$ benchmark (pool 962, test 240). R^2 at fixed budget; final R^2 and top-10 recall at budget 430; measurements to reach $R^2 = 0.85$.

strategy	R^2@100	R^2@250	final R^2	recall	labels→0.85
random	0.543	0.754	0.851	0.50	404
greedy	0.272	0.506	0.633	1.00	n/a
uncertainty	0.396	0.679	0.870	0.38	400
UCB	0.438	0.600	0.701	1.00	n/a
EI	0.379	0.664	0.776	0.98	n/a
hybrid	0.379	n/a	0.782	0.93	n/a

4 Baseline and prospective validation

A random forest trained on the published training split attains $R^2 = 0.942$, RMSE$= 0.328$ on the validation split (cf. the reported deep network $R^2 = 0.85$, RMSE$= 0.53$). Trained on all 1,202 measurements and applied to the four prospective ligands, it attains $R^2 = 0.687$, RMSE$= 0.639$, MAE$= 0.502$ log units.

Table 2: Active learning on the Lipophilicity benchmark (pool 3360, test 840), Morgan-fingerprint features. Final values at budget 715.

strategy	R^2@150	final R^2	recall
random	0.253	0.497	0.27
greedy	0.105	0.206	0.53
uncertainty	0.212	0.501	0.20
UCB	0.139	0.281	0.57
EI	0.143	0.309	0.53
hybrid	0.165	0.384	0.30

Table 3: Uncertainty calibration (f-element benchmark, 10 seeds). Empirical coverage of nominal-α intervals.

nominal α	uncalibrated	conformal
0.50	0.656	0.503
0.80	0.904	0.778
0.90	0.950	0.886
mean calib. error	0.103	0.013

5 Revision experiments

The following tables collect the full results of the leakage-controlled evaluation, the definition-resolved discovery metrics, the subgroup-resolved calibration, and the model/uncertainty-estimator comparison added in revision. Every value is generated from the released results files by `revision/make_tables.py`.

6 Reproduction

All code is released at `https://github.com/krishnatheaverage/logd-active-learning-benchmark` and archived on Zenodo. The original benchmark is reproduced by `src/`: `real_data.py` (baseline), `real_al.py` (f-element AL), `lipo_al.py` (Lipophilicity AL), `al_calibration.py` (calibration), `prospective.py` (prospective validation). The revision experiments are reproduced by `revision/`: `data_ext.py` (recovers ligand/source/lanthanide/condition group structure), `e1_splits.py` (leakage-controlled splits), `e2_ligand_batch.py` (ligand-batch

Table 4: Active learning with vs. without conformal calibration (final R^2 / top-10 recall). Acquisition performance is unchanged within noise.

strategy	uncalibrated	conformal
UCB	0.647 / 1.00	0.635 / 1.00
EI	0.696 / 0.98	0.709 / 0.98
hybrid	0.706 / 0.90	0.706 / 0.90

Table 5: Random-forest generalisation under increasingly stringent, leakage-controlled splits. Random row-level cross-validation is strongly optimistic; grouped splits that forbid a ligand, source, or structural family from appearing in both train and test collapse the estimate toward the mean. Mean±s.d. over 4 repeats of 5-fold grouped CV.

Split	R^2	RMSE	MAE
Random (row-level)	0.88±0.03	0.46	0.28
Ligand-disjoint	0.20±0.31	1.14	0.87
Source-disjoint	-0.03±0.80	1.22	0.98
Ligand-family (scaffold proxy)	-0.24±0.50	1.31	1.04
Prospective (4 new ligands)	0.70	0.63	0.49

acquisition), `e3_recall_metrics.py` (recall definitions and discovery metrics), `e4_calibration.py` (subgroup calibration and Mondrian), `e5_uq_baselines.py` (model/UQ comparison), with `make_figures.py` and `make_tables.py` regenerating every figure and table. The f-element source data are from Liu et al. (cited); the Lipophilicity set is from MoleculeNet.

Table 6: Discovery performance at the final budget under three definitions of a 'top-10 system' (condition-specific row, ligand–metal pair, unique ligand), plus simple regret (log units) and hit-rate above a practical threshold ($\log D \geq 1$, i.e. $D \geq 10$). The ranking of strategies inverts with the definition: exploitation dominates for rows, but broad sampling already recovers most top ligands. Means over six seeds.

Strategy	recall (row)	recall (lig–metal)	recall (ligand)	regret (log)	hit-rate ($\log D \geq 1$)
random	0.45	0.72	0.95	0.15	0.26
greedy	1.00	0.88	0.82	0.00	0.60
uncertainty	0.42	0.85	0.93	0.12	0.26
ucb	1.00	0.90	0.88	0.00	0.59
ei	0.93	0.90	0.97	0.02	0.47
hybrid	0.80	0.87	0.97	0.01	0.52

Table 7: Uncertainty calibration under a realistic ligand-disjoint split (nominal coverage 80%). Raw ensemble intervals under-cover; a single global split-conformal factor restores average coverage but leaves subgroup imbalance, which group-conditional (Mondrian) conformal reduces. Mean over eight seeds. For reference, on the easier row-level split the raw mean calibration error is 0.103 and conformal reduces it to 0.013.

Method	global	low $\log D$	mid $\log D$	high $\log D$
Raw	0.72	–	–	–
Global conformal	0.90	0.87	0.92	0.87
Mondrian conformal	0.87	0.96	0.82	0.88

Table 8: Model and uncertainty-estimator comparison on a ligand-disjoint split (train/calibration/test = 55/20/25 of ligands). Uncertainty-ranking quality is the Spearman correlation between predicted σ and realised absolute error; higher is better. Coverage is the empirical coverage of split-conformal 80% intervals. Mean over four seeds.

Model	R^2	conformal 80% cov.	rank quality $\rho(\lvert e\rvert, \sigma)$	enrich. factor
Random forest	0.15	0.91	0.22	7.5
Quantile RF	0.14	0.91	0.22	7.5
Gaussian process	-0.03	0.95	0.23	3.7
Gradient boosting	0.22	0.91	0.28	7.5
MC-dropout BNN	0.12	0.94	0.06	6.6